

# PUNIT MISHRA

Senior ML & AI Tech Lead | Agents, Inference & Multi-Tenant Platforms at Enterprise Scale

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## PROFILE

I like taking machine learning the whole distance — from rough prototype to something a real customer can lean on — and I’ve done it at enterprise scale for 14+ years. As Senior ML & AI Tech Lead in SAP CX Data & AI I *lead a team of thirteen* while staying hands-on: I **architect the systems and also write, deploy, and own them end to end**, from agent runtimes and LLM inference pipelines to **GPU serving on custom bare-metal H100 clusters**. Recently I brought **SAP’s Nova model into our natural-language-to-SQL stack** for cost control and model optionality, and architected a **hardened internal-auth contract** that closes a cross-tenant data-leak class. I care about the “so what”: automated onboarding over manual tenant setup, latency cut on our flagship retail deployment, security you can actually verify. Deep in **Python** and **Rust**, with multi-model serving, distributed inference, vector search, and **secure, ethical AI**.

## EXPERIENCE

|                                         |                    |
|-----------------------------------------|--------------------|
| <b>SAP</b>                              | Sep 2018 – Present |
| Senior ML & AI Tech Lead   CX Data & AI | San Ramon, CA      |

- | I brought **Nova**, SAP’s own code LLM, into our **natural-language-to-SQL** stack (S/4HANA + Ariba) as a first-class alternative to Claude — self-hosted serving, a retriever-reranker, and seven benchmark lanes — **cutting dependence on external frontier models** and lifting decision-match **59.2% → 64.8%**; NL-to-SQL ships in SAP’s **2026 public and private cloud releases**.
- | I architected the **hardened internal-auth contract** (ADR 0021) — a constant-time shared-secret check before any tenant header is trusted, canonical tenant resolution, and **IAS alongside XSUAA** — shipped backward-compatibly across four services to **close a real cross-tenant data-leak class**.
- | I turned APEX from manual, XSUAA-only tenancy into **direct app tenancy auto-provisioned via BTP UCL formations, enabling self-service customer onboarding** — and serve as **forward-deployed engineer** bringing up our first production enterprise customer, **a Fortune 500 retailer**.
- | I rewrote the legacy Shopping Assistant’s **vision** (Vision-DNA, object detection, visual search) into first-class agent services behind a unified **Agent Gateway** and an **MCP Hub**, turning dead-end legacy code into reusable, discoverable services.
- | I **lead a team of thirteen** and set technical direction through the **ADRs** I author, our SDLC and engineering standards, and **AI ethics & compliance** as an **SAP Security Expert**.
- | I built the **APEX Control Panel**, a Next.js provisioning UI (IAS OIDC PKCE + server-side BFF) — onboarding, formations, agent & data-ingestion config, playground, sessions, handoff — and folded the AutoSQL eval dashboard into a **HANA-backed Nova dashboard on Gardener**, giving the team self-service operational visibility and a reusable FDE UI scaffold.
- | I run **multi-model serving** — self-hosted Nova alongside Claude, Qwen, and Granite-class models — on custom bare-metal H100 clusters, and **introduced Ray** plus automated eval harnesses so we choose models on measurable accuracy, not intuition.
- | I cut **“ask a product” latency** on a **Fortune 500 retailer’s** in-store concierge from **~25s to ~10s** and added grounded follow-ups that stop product drift — real customer-facing speed and quality wins on our flagship deployment.
- | I architect **APEX**, our domain-agnostic Digital Service Agents platform, writing the performance-critical hot paths in **Rust** so one agent runtime is fast and cheap enough to reuse across products instead of rebuilding per use case.

|                                                               |                                       |
|---------------------------------------------------------------|---------------------------------------|
| <b>CALLIDUSCLOUD</b>                                          | Feb 2014 – Sep 2018                   |
| Senior Software Engineer   Thunderbridge AI / SAP Sales Cloud | SF Bay Area (acquired by SAP, \$2.4B) |

- | Designed and built the **Thunderbridge AI** backend in Python on **Dataiku**, with multi-tenancy running every customer workload for SAP Sales Cloud optimization.
- | Ported the full Dataiku ML stack to **GKE on-prem (Google Anthos)** with Istio, and built the CI/CD backbone — Jenkins on Kubernetes, Kaniko image builds, and Artifactory distribution.
- | **Built developer-productivity tooling** — automated pull-request merges with Palantir **Bulldozer** and **Policy-Bot** on Kubernetes, wired into SAP GitHub Enterprise via OAuth to capture all Git events — cutting manual merge overhead across the team.

- | Engineered multi-partition data resiliency for ML across **Oracle, Vertica, SAP HANA, and HDFS** (Spark); shipped inference services for CPQ on Dataiku's API node.
- | Integrated SAP security scans (**Coverity, Black Duck, Fortify**) into Jenkins per commit. As Senior Automation Engineer (earlier), built TDD / Scala ETL validation and resilient Selenium/Java and Node.js UI test frameworks over a self-maintaining Selenium Grid.

## IBM

Automation Systems Engineer

Oct 2012 – Dec 2013  
Dallas–Fort Worth, TX

- | Built a reporting and **data-validation automation framework** for a major US bank.
- | Automated VM creation/configuration via a WCF SOAP service (IIS); parallelized test execution and scheduled runs across cloud machines with varied OS/browser configs.
- | Developed reusable **Selenium / Java** test APIs (JUnit, TestNG) adopted across the firm.

## UC BERKELEY LAW IST

Web Developer

Sep – Dec 2011  
Berkeley, CA

- | Built an online teaching-evaluation system for the Law School on a PHP / Oracle 10g backend, integrated with **CalNet** (Berkeley authentication) for end-of-course student evaluations.

## LAWPIVOT INC. (GOOGLE STARTUP LABS)

Software Engineering Intern

2009 – 2011  
Mountain View / Fremont, CA

- | Built user-to-user interaction features on the **Django** backend and an internal analytics tool visualizing user data (logins, time on site) used by the sales team.
- | Earlier, automated testing with Selenium IDE and tracked defects; worked with the founding team on wireframes and AWS server configuration.

## SELECTED PROJECTS & OPEN SOURCE

- | **CoViber** — local-first memory for LLMs over **MCP**: a continuously-updated work graph, urgency-triage queue, and local semantic memory, entirely on-device (no cloud egress). *Python*.
- | **Vaultic** — hardware-backed (FIDO2 / YubiKey) password manager in **Rust**; one on-disk vault format across CLI, TUI, daemon, GUI, and an MCP server. *Homebrew tap*.
- | **Shield AI** — AI-powered DNS filtering with real-time **ML threat scoring**; **Rust** + TypeScript, ~15 MB resident and <1 ms lookups at 100K+ QPS. *Live demo + macOS/iOS DNS profile*.
- | **Railroad Arcade** — full-stack IoT (Next.js + **Rust** on a Raspberry Pi) driving a real HO-scale model railroad remotely, with demo and live-control modes.
- | **Pi Race** — zero-dependency **Rust** CLI benchmarking four classic  $\pi$  algorithms (Ramanujan, Euler/Basel, Wallis, Leibniz) on accuracy vs. speed; CI cross-compiles Linux / macOS / Windows and posts weekly timing tables.

## SKILLS

|                   |                                                                                                                                                                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI & AGENTS       | Agent orchestration (LangGraph, LangChain), <b>Agent Gateway, MCP Hub</b> (registry / discovery), LLM serving, fine-tuning & evaluation, NL-to-SQL, RAG, retriever-reranker, PyTorch, OpenAI / CLIP, automated eval harnesses & benchmark lanes |
| GPU & SERVING     | <b>Ray</b> (distributed inference / eval engines), custom bare-metal H100 clusters, <b>multi-model serving</b> (self-hosted Nova + Claude / Qwen / Granite-class), self-hosted LLM / embedding serving, Knative / KServe                        |
| INFRA & CLOUD     | Kubernetes, Kyma / Gardener, Istio, Argo / GitOps, Jenkins, Docker, CI/CD; vector DBs (Qdrant, FAISS), Redis; Azure (AKS), GCP (GKE / Anthos)                                                                                                   |
| PLATFORM & DATA   | gRPC / protobuf, IAS / XSUAA (OIDC / PKCE), CAP / cds-mtxs, BTP UCL formations; PostgreSQL, SAP HANA, Oracle, Vertica, HDFS / Spark, Kafka                                                                                                      |
| LANGUAGES         | <b>Python</b> (FastAPI, Django), <b>Rust</b> , Java, Scala, TypeScript / Next.js                                                                                                                                                                |
| DEV PRODUCTIVITY  | GitHub Enterprise automation (Bulldozer, Policy-Bot), Artifactory, Kaniko, ADRs, ephemeral PR environments                                                                                                                                      |
| SECURITY & ETHICS | SAP Security Expert; internal-auth / identity design, threat modeling, SAST / SCA (Coverity, Black Duck, Fortify); AI ethics, safety & compliance                                                                                               |

## EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY — B.S. Electrical Engineering & Computer Science

2012